

NU CSE 1st semester questions 2019

1)EEC

2)Calculus

3)physics

4)SPL

5)English

prepared by



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A decorative graphic on the left side of the cover, featuring a black background with a grid of white dots and blue lines representing circuit traces.

COMPUTER SCIENCE AND ENGINEERING

NATIONAL UNIVERSITY 2019 1ST SEMESTER QUESTIONS

B. Sc (HONS.) IN CSE, PART-I, FIRST SEMESTER EXAMINATION, 2019

[According to the New Syllabus]

CSE-510203

(Electrical and Electronic Circuit)

Time—3 hours

Full marks—80

[N.B.—The figures in the right margin indicate full marks. Answer any four questions. Each part of a question must be answered sequentially.]

Marks

- | | | | |
|----|-----|--|---|
| 1. | (a) | What is impedance? Find the impedance of a series RLC circuit. | 5 |
| | (b) | Define Power factor and Q-factor of a coil. | 4 |
| | (c) | A coil takes a current of 6A when connected to a 24V d.c supply. To obtain the same current with a 50 Hz a.c. supply, the voltage required was 30V. Calculate (i) the impedance of the coil (ii) the power factor of the coil. | 6 |
| | (d) | Show that the voltage of a RL circuit leads with current of a AC signal. | 5 |
| 2. | (a) | State and explain Kirchhoff's current and voltage law. | 6 |
| | (b) | What is biasing? Why biasing is necessary for a transistor circuit? | 4 |
| | (c) | What is semiconductor? Explain the effect of temperature on semiconductor. | 4 |
| | (d) | Describe the classification of semiconductor. Explain the n-type semiconductor with energy band diagram. | 6 |
| 3. | (a) | Describe the operation of the P-N junction diode with V-I characteristics. | 6 |
| | (b) | Draw the full wave bridge rectifier circuit and describe its operation. | 6 |
| | (c) | What are the advantages of a full wave rectifier over a half wave rectifier? | 4 |
| | (d) | A germanium diode draws 40 mA with a forward bias of 0.25V. The junction is at room temperature of 293°K. Calculate the reverse saturation current of the diode, | 4 |
| 4. | (a) | What is BJT? Explain the working principle of npn transistor. | 6 |
| | (b) | Describe the input and output characteristics of Common Base connection. | 6 |
| | (c) | Explain the operation of zener diode as a voltage regulator. | 4 |
| | (d) | Draw load line and determine the operating point of the following circuit (Assume the transistor to be at silicon). | 4 |

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- | | Marks |
|---|--------|
| 5. (a) What is FET? Describe the operation of FET as a switch with appropriate circuit. | 5 |
| (b) Write down the characteristic of an ideal operational amplifier. | 4 |
| (c) Show that Op-Amp works as the following :
(i) Inverting amplifier (ii) Integrator (iii) Summing amplifier. | 6 |
| (d) Define negative feedback. Explain how negative feedback reduces noise and non-linear distortion. | 5 |
| 6. Write short notes (any four) : | 5×4=20 |
| (a) Norton's Theorem | |
| (b) Photo Diode | |
| (c) <i>r.m.s</i> value of average value of an ac signal | |
| (d) Bandwidth | |
| (e) Voltage follower | |
| (f) MOSFET. | |

B. Sc (HONS.) IN CSE, PART-I, FIRST SEMESTER EXAMINATION, 2019

[According to the New Syllabus]

CSE-510205

(Calculus)

Time—3 hours

Full marks—80

[N.B.—The figures in the right margin indicate full marks. Answer any four questions.]

Group A—Differential Calculus

Marks

1. (a) Define domain and range with example. 4
- (b) Sketch the graph of the function : 8
 $f(x) = |x - 1| + |x + 3|$
 Also find the domain and range. 4
- (c) A function $f(x)$ defined as follows :

$$f(x) = \begin{cases} 5x - 4 & \text{when } 0 < x \leq 1 \\ 4x^2 - 3x & \text{" } 1 < x < 2 \\ 3x + 4 & \text{" } x \geq 2 \end{cases}$$

 Test the continuity at $x = 1$ and $x = 2$. 4
- (d) By (σ, ϵ) definition, prove that :
 $\lim_{x \rightarrow 3} \frac{2x^2 - 18}{x - 3} = 12$. 4
2. (a) Find $\frac{dy}{dx}$ (any three) : $4 \times 3 = 12$
 (i) $y = \tan^{-1} \left(\frac{1 + \tan x}{1 - \tan x} \right)$
 (ii) $y = x \cos^{-1} x + (\sin x) \ln x$
 (iii) $x + y = \sin^{-1} \left(\frac{y}{x} \right)$
 (iv) $x = a \cos^3 t, y = a \sin^3 t$
- (b) State Leibnitz's theorem. If $x = \sin \left(\frac{1}{m} \ln y \right)$ then show that— $2 + 6 = 8$
 $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (n^2 + m^2)y_n = 0$.

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3. (a) State and prove Rolle's theorem.

6

(b) Evaluate : $\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x - \sin x}$

4

- (c) Discuss the application of Rolle's theorem to the function $f(x) = x^2$ in $(-1, 1)$.

5

- (d) A farmer with a field adjacent to a straight river wishes to fence off a rectangular area for grazing. If no fence is needed along the river and he has 2400m of fencing. What should be the dimensions of the field in order that it has a maximum area?

5

Group B—Integral Calculus

4. Evaluate (any five) :

4×5=20

(i) $\int \sin^5 x \cos^3 x \, dx$

(ii) $\int e^x \frac{(2 + \sin 2x)}{1 + \cos 2x} \, dx$

(iii) $\int \frac{7x - 9}{x^2 - 2x + 35} \, dx$

(iv) $\int \frac{dx}{(x^2 + 4x + 5)^2}$

(v) $\int \frac{dx}{5 + 4 \cos x}$

(vi) $\int \frac{e^{m \tan^{-1} x}}{1 + x^2} \, dx$

5. (a) Define gamma function. Prove that :

2+6=8

$$\int_0^{\frac{\pi}{2}} \sin^m x \cos^n x \, dx = \frac{\Gamma\left(\frac{m+1}{2}\right) \Gamma\left(\frac{n+1}{2}\right)}{2\Gamma\left(\frac{m+n+2}{2}\right)}$$

(b) Evaluate (any three) :

Marks

4×3=12

$$(i) \int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin x} \, dx}{\sqrt{\sin x} + \sqrt{\cos x}}$$

$$(ii) \int_0^{\infty} \frac{(\tan^{-1} x)^2}{1+x^2} dx$$

$$(iii) \int_0^{\frac{\pi}{2}} \frac{dx}{1+\cot x}$$

$$(iv) \int_0^{\frac{\pi}{2}} \sin^6 x \cos^3 x \, dx$$

6. (a) Find the length of the arc of the parabola $y^2 = 16x$ between the vertex and one end of the latus rectum. 6

(b) Show that the area between the parabola $y^2 = 4x$ and the straight line $y = 2x - 4$ is 9. 7

(c) Find the volume of the solid generated by revolving the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ about x-axis. 7

B. Sc (HONS.) IN CSE, PART-I, FIRST SEMESTER EXAMINATION, 2019

[According to the New Syllabus]

CSE-510207

(Physics)

Time—3 hours

Full marks—80

[N.B.—The figures in the right margin indicate full marks. Answer any four questions.]

- | | | Marks |
|----|--|-------|
| 1. | (a) State and explain Coulomb's law. Write Coulomb's law in vector form. | 6 |
| | (b) What is electric field and electric field strength? Explain how electric field is calculated. | 3+4=7 |
| | (c) What is electric dipole and electric dipole moment? Calculate the E due to the charges of an electric field at a distance r along the perpendicular bisector of the line joining the charges. | 2+5=7 |
| 2. | (a) State and explain Gauss's law. | 4 |
| | (b) Derive Coulomb's law from Gauss's law. | 5 |
| | (c) What is potential? Calculate the potential due to a point charge. | 1+5=6 |
| | (d) Calculate the capacitance of a parallel plate capacitor. | 5 |
| 3. | (a) Write down the comparisons among Dia, Para and Ferro-magnetism. | 5 |
| | (b) What is self and mutual inductance? Define their co-efficients. | 5 |
| | (c) Derive the expression for the rising current and define time-constant of an R-L circuit with DC voltage. | 6 |
| | (d) A series R-L circuit has $R = 100 \text{ K}\Omega$ and $L = 10 \text{ mH}$. If a DC voltage source is connected to its circuits, determine the value of current as the % of the maximum current. | 4 |
| 4. | (a) Define negative field vector and explain the concept of line of induction. What happens when a positive test charge $+q$ moves through a uniform site of magnetic field with constant velocity? | 6 |
| | (b) What is Lorentz force? Explain. | 4 |
| | (c) Derive an expression for magnetic induction $\vec{B}$ at a distance ' r ' from the center of a long cylindrical wire of radius R , where $r < R$. Assume the wire carries i_0 current distributed uniformly over the cross section of the wire. | 5 |
| | (d) Two parallel conductors are ' d ' distance apart. If both of them are of ' l ' length and carries i_0 current, then find the nature and magnitude of force that exists between the two conductors. | 5 |

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5. (a) Mention and explain Biot-Savart's law.
- (b) What is capacitance? Find the equivalent capacitance when three $100\mu\text{F}$ capacitors are connected in series and then in parallel.
- (c) What is self inductance? Derive an expression of potential energy stored in magnetic field in a series RL circuit.
- (d) Explain the working principle of a moving coil galvanometer with required figure.

6. Write short notes on any four :

5x4

- (a) Co-efficient of magnetic coupling
- (b) Hysteresis and eddy-current loss
- (c) Resonance of series RLC circuit
- (d) Wattmeter
- (e) E.M. F.
- (f) Magnetic field of solenoid

B. Sc (HONS.) IN CSE, PART-I, FIRST SEMESTER EXAMINATION, 2019

[According to the New Syllabus]

CSE-510201

(Structured Programming Language)

Time—3 hours

Full marks—80

[N.B.—The figures in the right margin indicate full marks. Answer any four questions.]

- | | | Marks |
|----|--|-------|
| 1. | (a) What is structured programming language? Write down the characteristics of structured programming language. | 5 |
| | (b) Define algorithm and flow-chart. Describe the symbols used in flow-chart. | 5 |
| | (c) What is pre-processor directive and header file? Give example. | 5 |
| | (d) Write down the algorithm to calculate area and <u>circumference</u> of a circle. | 5 |
| 2. | (a) Define token, keyword, identifier, variable and constant. | 5 |
| | (b) What is symbolic constant? Write down the rules for declaring symbolic constant. | 5 |
| | (c) What are the different types of operator used in C language? Explain increment and decrement operator with example. | 5 |
| | (d) Write a C program to determine the largest value from three numbers. | 5 |
| 3. | (a) Define loop and looping process. Explain entry controlled and exit controlled loops. | 6 |
| | (b) What are the differences between while and do.....while loop? | 3 |
| | (c) Write down the general form and flow-chart of the following statement :
(i) elseif ladder (ii) SWITCH (iii) FOR | 6 |
| | (d) Write a program in C to find the factorial of an integer. | 5 |
| 4. | (a) What is an array? Write the advantages and disadvantages of array. | 1+4=5 |
| | (b) Mention some string handling functions and describe them with example. | 5 |
| | (c) What is pointer? Write down the merits and demerits of using pointer with respect to array. | 5 |
| | (d) Write a C program to multiply two matrices. | 5 |

[Please turn over]

- | | | Marks |
|----|---|-------|
| 5. | (a) Define user-defined function and library function with example. | 4 |
| | (b) What do you mean by actual and formal parameter? Explain with an example. | 5 |
| | (c) Define structure and union. Write down the differences between structure and array. | 6 |
| | (d) Write a C program to calculate the sum of the following series :
$1 + 2^2 + 3^3 + 4^4 + \dots + 50^{50}$. | 5 |
| 6. | (a) What is file? Describe different file opening modes. | 1+4=5 |
| | (b) What do you mean by dynamic memory allocation? What are the advantages of using linked list over array? | 5 |
| | (c) Write down the differences between printf() and fprintf(). | 4 |
| | (d) Write a program to append the contents of one file to another file. | 6 |

B. Sc (HONS.) IN CSE, PART-I, FIRST SEMESTER EXAMINATION, 2019

[According to the New Syllabus]

GED-510209

(English)

Time—3 hours

Full marks—80

[N.B.—The figures in the right margin indicate full marks. Answer any four questions. Each part of a question must be answered sequentially.]

1. Read the following passage and answer the questions below :

The great advantage of early rising is the good start it gives us in our day's work. The early riser does a large amount of work before other men have got out of bed. In the early morning, the mind is fresh and there are few sounds or other distractions so that work done at that time is generally well done. In many causes, the early riser also finds time to take some exercise in the fresh air of the morning and this exercise supplies him with a fund of energy that may last until the evening. By rising so early he knows that he has plenty of time to do thoroughly all the works he can be expected to do, and is not tempted to hurry over any part of these. By finishing all the works on time, he enjoys a long interval of rest in the evening before the timely hour when he goes to bed. He goes to sleep several hours before midnight at the time when sleep is most refreshing and after a sound rest at night, rises early next morning having good health and spirits for the labours of a new day. It is very plain that such a life as this is far more conducive to health than that of the man who shortens his waking hours by rising late and so can afford in the course of the day little leisure for necessary rest. Anyone who lies in bed late, must, if he wishes to do a full day's work, go on working to a correspondingly late hour and deny himself the hour or two of evening exercise that he ought to take for the benefits of his health.

In spite of all his efforts, he will not probably produce as good results as the early riser, because he misses the best working hours of the day.

Marks

Questions :

- | | |
|---|---|
| (a) (i) Why does the late riser fail to produce good results? | 5 |
| (ii) What does a person know by rising early in the morning? | |
| (b) Write down the meaning of the following words and make sentences with them (any five) : | 5 |
| advantage, supply, plenty, interval, generally, conducive. | |
| (c) What is the main idea and what are the supporting ideas of the passage? | 5 |
| (d) Write a summary of the passage. | 5 |

[Please turn over

2. (a) Frame WH-questions from the following sentences : 5

- (i) The TV set cost men Tk. 40,000.
- (ii) He has been living in Dhaka for five years.
- (iii) The mangoes taste sour.
- (iv) You must respect your teachers.
- (v) He went out to fetch water.

(b) Use the right form of verbs in the following sentences : 5

- (i) The bank (locate) at Agrabad.
- (ii) The mechanic (repair) the pump and it is working now.
- (iii) We (have) our breakfast just now.
- (iv) Football (play) between two teams.
- (v) They (live) in Dhaka for the last ten years.

(c) Correct the following sentences : 5

- (i) He hanged the shirt on the wall.
- (ii) It is a true fact.
- (iii) I shall avail the opportunity.
- (iv) The Cholera has broken out in the village.
- (v) The wages of sin are death.

(d) Combine each of the following groups of sentences into one sentence : 5

- (i) You may like it. You may not like it. I shall give you good advice. It is my duty to do so.
- (ii) I was returning home. I saw a man. He looked very happy. He was singing a song.

3. (a) Amplify the followings : 10

Empty vessel sounds much.

(b) Translate into English : 10

আমরা প্রতিদিন সংবাদপত্রের পাতায় সন্ত্রাসী কর্মকাণ্ডের খবর পড়ি। দিনের পর দিন ইহা বৃদ্ধি পাচ্ছে। ইহা আমাদের জাতীয় জীবনে অন্যতম একটি সমস্যা। আমরা সুন্দর ও সুস্থ জীবন যাপন করিতে চাই। মনে রাখিতে হইবে একটি সন্ত্রাসী কর্মকাণ্ড আরেকটি সন্ত্রাসী কর্মকাণ্ডকে উৎসাহিত করে। যারা সন্ত্রাসীকে আশ্রয় দেয় তাদেরকে এই পথ পরিহার করা উচিত। আসুন আমরা সকলে মিলে সন্ত্রাস ও সন্ত্রাসীদের মোকাবেলা করি। ইহা আমাদের মৌলিক দায়িত্ব।

4. Write an essay on any **one** of the following :
- (a) National Calamities in Bangladesh;
 - (b) Duties to parents;
 - (c) Students and Social service.
5. (a) Write a dialogue between two friends about a visit to a place of historical interest. 10
- (b) Suppose, you are a reporter of a national daily. Now, write a report on the water logging in the city streets. 10
6. (a) Write a paragraph on any **one** of the following : 10
- (i) Greenhouse effect
 - (ii) Truthfulness
- (b) Write an application to the principal of your college seeking permission to go on a study tour. 10